

Minjae Kwen

Curriculum Vitae

Quantum Science and Engineering, Harvard University,
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Research Interests

Quantum Information & Computation

- Analog Quantum Simulation using Rydberg Atom Arrays
- Quantum State Preparation, Error Correction (QEC), and Machine Learning (QML)

Many-Body Physics & Dynamics

- Dynamics of Open and Complex Quantum Systems
- Quantum Many-Body Theory at the Chemistry–Physics Interface

Computational Chemistry & Materials

- Nonadiabatic Dynamics (Surface Hopping) / *Ab-initio* Electronic Structure Calculations

Education

Harvard University, Cambridge, MA

2025 – Present

Ph.D. Student in Quantum Science and Engineering

KAIST, Daejeon

2019 – 2025

B.S in Chemistry (Minor: Material Science)

- Valedictorian (GPA: 4.18/4.3, *Summa Cum Laude*)
- Military Service, Alternative: May. 2022 – Feb. 2024
- KAIST-IIT Madras Joint Research Challenge (Feb. 2020)
- UC Berkeley Summer Sessions (Jun.–Aug. 2019)

Daegu Science High School, Daegu

2016 – 2019

High school for the gifted in science and mathematics

Skills

Programming

Python, C, Bash

Simulation Tools

VASP, LAMMPS, Quantum ESPRESSO

Scientific Libraries

QuTiP, QuSpin, JAX, PyTorch, NumPy, CuPy, NequIP

Languages

Korean (native), English (fluent)

Publications

Journal Articles

1. Baik, Y., Kwen, M. *et al.* Splitting of hydrogen atoms into proton–electron pairs at BaO–Ru interfaces for promoting ammonia synthesis under mild conditions. *Journal of the American Chemical Society* 145.20 (2023): 11364–11374. (Doi: [10.1021/jacs.3c02529](https://doi.org/10.1021/jacs.3c02529).)

Conference Presentations

1. Kwen, M. *et al.* (2024). Time-domain ab initio analysis of facet-dependent carrier dynamics in Cuprous oxide, ISTCP 2024, Poster ([link](#))

Research Experience

Yelin Group (Harvard)

Feb. 2026 – Present

Dissipative state preparation for analog quantum simulator using Rydberg atom arrays
(Advisor: Susanne Yelin)

Lee Group (Harvard)

Sep. 2025 – Jan. 2026

Machine Learning prediction of phonon dispersion in crystal (Advisor: Joonho Lee)

M-design Lab (KAIST)

Sep. 2021 – Aug. 2025

Facet-dependent carrier dynamics; proton–electron interface storage; and DFT screening for NO electroreduction (Advisor: Hyungjun Kim)

Honors and Awards

Overseas PhD Scholarship

2024–Present

Korea Foundation for Advanced Studies (KFAS)

Korean Chemical Society Presidential Award

2025

Korean Chemical Society (KCS)

Korean Presidential Science Scholarship, Chemistry

2019–2025

Korea Student Aid Foundation (KOSAF)

KAIST Presidential Fellowship (KPF)

2019–2025

Global Leadership Center, KAIST